

		$V = 9.4 \text{ V}$ (2 s.f. scores 2/3)		A1	[3]
			Total		[7]
8	(a)	S shown at the peak		B1	[1]
	(b)	(i) Kr and U on right of peak in correct relative positions		B1	[1]
		(ii)1 binding energy of U-235 = $2.8649 \times 10^{-10} \text{ J}$			
		binding energy of Ba-144 = $1.9211 \times 10^{-10} \text{ J}$			
		binding energy of Kr-90 = $1.2478 \times 10^{-10} \text{ J}$		C2	
		energy release = $3.04 \times 10^{-11} \text{ J}$ (-1 if 1 or 2 s.f.)		A1	[3]
		2 $E = mc^2$		C1	
		$m = (3.04 \times 10^{-11}) / (3.0 \times 10^8)^2 = 3.38 \times 10^{-28} \text{ kg}$ (ignore s.f.)		A1	[2]
	(iii)	e.g. neutrons are single particles,			
		neutrons have no binding energy per nucleon		B1	[1]
			Total		[8]